

## Arrival choices

Previous lesson reminder:

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

### 1 What was recorded at Coningsby on 19 July 2022?

Support: Use the reminder strip or talk it through with an adult.

My response

---

---

---

### 2 Which meaning fits tectonic?

Support: Choose: Related to movement of Earth's plates OR People or things in harm's way.

My response

---

---

---

### 3 Which units belong on the temperature axis?

Support: Use the model evidence anchor dataset or read one relevant card together.

My response

---

---

---

### 4 Can selected station maxima establish the temperature of every street?

Support: Give a reason when ready.

My response

---

---

---

Stretch: use a precise example or explain which detail supports your answer.

## Evidence cards

Read, point or listen. Keep the source label with any detail you use.

### A Japan 11 March 2011

A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Source: USGS event record, official20110311054624120\_30

Limit: Magnitude measures the earthquake; it is not a direct count of people affected.

### B Effects and responses

Shaking and coastal flooding damaged places and disrupted people's lives. Earthquake-resistant design addresses shaking; warnings, evacuation routes and higher ground address tsunami exposure.

Source: Authored explanation of the case and risk-management principles

Limit: A sea wall or warning cannot eliminate all risk. Do not judge people without considering access and warning time.

### C Heat and unequal risk

On 19 July 2022 the Met Office recorded 40.3°C at Coningsby, 40.2°C at St James's Park and 39.6°C at Topcliffe. Outdoor work increases exposure; difficulty accessing cool spaces can increase vulnerability. Shade and accessible cool spaces can reduce harm.

Source: Met Office July 2022 report, station table; risk explanation authored for teaching

Limit: This explains risk factors, not a medical assessment of an individual.

### D Reading a hazard feed

A USGS event row can include UTC time, place, magnitude and depth in kilometres. A short feed is a dated snapshot. It cannot predict the next earthquake.

Source: USGS earthquake feed documentation

Limit: Small samples, reporting thresholds and later revisions affect patterns.

## Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:  
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

---

---

---

# Data for the enquiry

Selected station maxima on 19 July 2022

| Station         | Maximum °C |
|-----------------|------------|
| Coningsby       | 40.3       |
| St James's Park | 40.2       |
| Topcliffe       | 39.6       |
| Sheffield       | 39.4       |

Units: °C

Source: Met Office July 2022 heatwave report, table on page 8

Limit: Four selected station maxima are not a national average or a long-term climate trend. The report marked values provisional at publication.

My labelled graph or data notes

---

---

---

## Shared investigation

1. Transfer the selected heatwave station values into the spreadsheet.

2. Choose a bar chart, label units and check the plotted values.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

Use the evidence cards and any visual sheet. Follow the two shared steps above, then record one decision.

Our explanation

---

---

---

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining.

Further challenge: Explain why a bar chart of selected station maxima cannot show an annual climate trend.

## Supported independent task

Complete this route only. Analyse hazard data in spreadsheets and produce a digital report.

1. Enter or check the case-study data and create a labelled graph.
2. Write a digital report paragraph describing a pattern with values.
3. State a limitation of the selected stations and data.

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining. Complete the organiser one space at a time. The highest value in this selection is \_\_\_\_ at \_\_\_\_.  
The range is \_\_\_\_\_. This sample cannot establish \_\_\_\_\_.

Data source and date or capture time

---

---

---

Variables units and graph labels

---

---

---

A pattern supported by values

---

---

---

What the data cannot establish

---

---

---

An adult may read or scribe your exact response. They should not choose the answer.

## Standard independent task

Complete this route only. Analyse hazard data in spreadsheets and produce a digital report.

1. Enter or check the case-study data and create a labelled graph.
2. Write a digital report paragraph describing a pattern with values.
3. State a limitation of the selected stations and data.

Support: The highest value in this selection is \_\_\_\_ at \_\_\_\_\_. The range is \_\_\_\_\_. This sample cannot establish \_\_\_\_\_.

Data source and date or capture time

---

---

---

Variables units and graph labels

---

---

---

A pattern supported by values

---

---

---

What the data cannot establish

---

---

---

Check the required evidence and revise one detail.

## Stretch independent task

Complete this route only. Analyse hazard data in spreadsheets and produce a digital report.

1. Enter or check the case-study data and create a labelled graph.
2. Write a digital report paragraph describing a pattern with values.
3. State a limitation of the selected stations and data.
4. Explain why a bar chart of selected station maxima cannot show an annual climate trend.

Support: Choose the evidence independently. Use the frame only if it helps.

Data source and date or capture time

---

---

---

Variables units and graph labels

---

---

---

A pattern supported by values

---

---

---

What the data cannot establish

---

---

---

Explain why a bar chart of selected station maxima cannot show an annual climate trend.

## Exit ticket

Choose a response mode. You may give your answers privately to an adult.

### Which units belong on the temperature axis?

My response

---

---

---

### Can selected station maxima establish the temperature of every street?

My response

---

---

---

Support: The highest value in this selection is \_\_\_\_ at \_\_\_\_\_. The range is \_\_\_\_\_. This sample cannot establish \_\_\_\_\_.

Further challenge: Explain why a bar chart of selected station maxima cannot show an annual climate trend.

One thing I changed or want to check

---

---

---